

ATM

An Automated Teller Machine Using Verilog HDL

PROJECT TEAM

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Verilog HDL Based Digital Design Project

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ABSTRACT

The Automated Teller Machine (ATM) is a digital system designed to provide basic banking services without direct assistance from a bank employee. This project presents the design and implementation of an ATM controller using Verilog Hardware Description Language (HDL).

The system supports multiple user operations such as card identification, PIN verification, balance enquiry, cash withdrawal, deposit, PIN change, and card ejection. The system maintains separate account information for three registered cards and provides security through PIN authentication and limited PIN attempts.

The ATM controller is designed using a Finite State Machine (FSM), where each state represents a particular stage of the ATM operation. The design is verified through RTL simulation and a testbench to observe the correct operation of different transactions.

1. INTRODUCTION

An Automated Teller Machine is an electronic banking system that allows customers to perform basic banking operations using a card and a Personal Identification Number (PIN).

In this project, an ATM controller is designed using Verilog HDL. The controller receives the card information and PIN from the user and provides access to different banking operations after successful authentication.

The main purpose of the project is to understand how a real-world banking system can be represented as a digital control system. The complete operation is controlled using a Finite State Machine, which provides an organized sequence of states for card insertion, PIN verification, transaction selection, transaction processing, and card ejection.

The design also includes security features such as PIN verification and blocking of a card after three consecutive incorrect PIN attempts. The project demonstrates the application of digital design concepts, FSM-based control, registers, counters, and RTL simulation in a practical system.

2. PROJECT OVERVIEW

The proposed ATM system is a Verilog HDL based digital controller that models the basic operations of an Automated Teller Machine.

The system is designed to handle three registered ATM cards. Each card is associated with a unique card number, PIN, and account balance. When a card is inserted, the controller identifies the card and proceeds to PIN verification.

After successful authentication, the user can select different banking operations. The available operations include:

- Balance enquiry
- Cash withdrawal
- Deposit
- PIN change
- Card ejection

For security, the system allows a maximum of three incorrect PIN attempts. After three consecutive incorrect attempts, the corresponding card is blocked for a temporary period.

The complete control operation is implemented using a Finite State Machine (FSM). Each state performs a specific function and the controller moves between states according to the inputs received from the user.

3. OBJECTIVES

The main objectives of the ATM project are:

- To design an ATM controller using Verilog HDL.
- To understand the working of an ATM system using digital logic.
- To implement card identification and PIN authentication.
- To provide basic banking operations such as withdrawal, deposit, and balance enquiry.
- To implement PIN changing functionality.
- To provide secure access by limiting incorrect PIN attempts.
- To implement card blocking after three incorrect PIN attempts.
- To design the control system using a Finite State Machine.
- To verify the functionality of the design using RTL simulation and a testbench.
- To understand the practical application of FSMs in digital system design.

Project Overview and Objectives

4. SYSTEM ARCHITECTURE

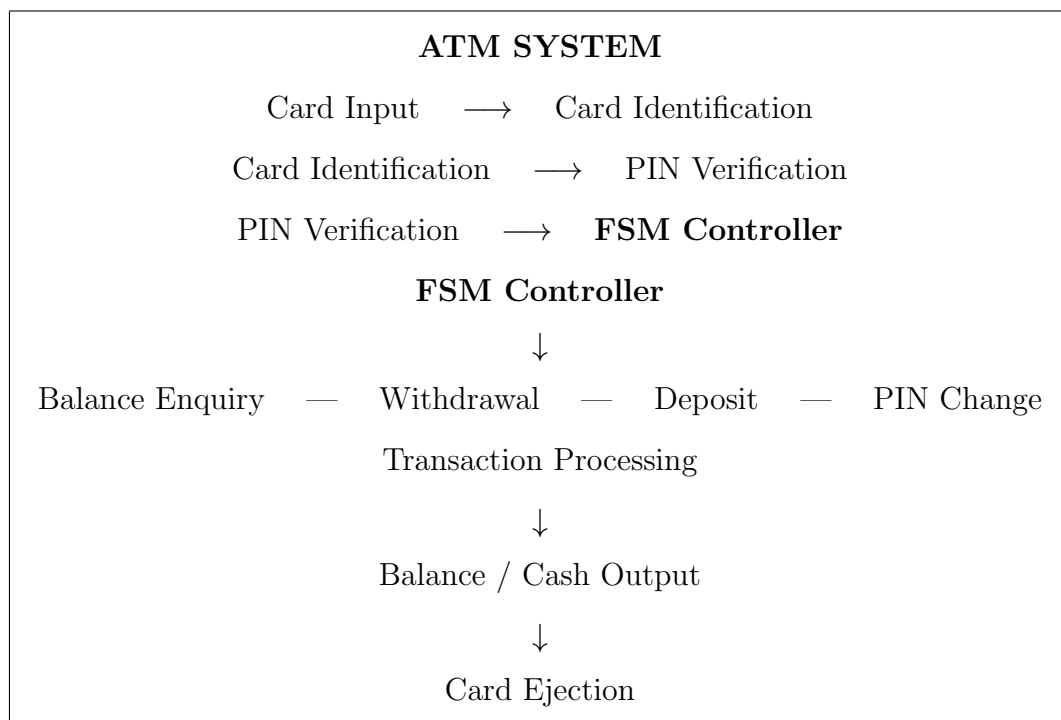
The ATM controller is organized into different functional sections. These sections work together to control the complete ATM operation.

The major parts of the system are:

- Card Identification
- PIN Verification
- Transaction Selection
- Balance Management
- Withdrawal
- Deposit
- PIN Change
- Security and Attempt Control
- Card Ejection
- Finite State Machine Controller

The inputs from the user are processed by the FSM controller. Based on the current state and input conditions, the controller performs the required operation and produces the corresponding outputs.

Overall Block Diagram



figureOverall Architecture of the ATM Controller

Working Principle:

The user first provides the card information. The controller checks whether the card belongs to one of the registered accounts. If the card is valid, the system proceeds to PIN verification.

After successful PIN verification, the user can select the required transaction. The FSM controller then moves to the appropriate state and performs the selected operation.

The account balance is updated after deposit or successful withdrawal. The system can also display the current balance and allows the user to change the PIN. Finally, the card can be ejected from the system.

6. SYSTEM OPERATION

The ATM controller operates in a sequence of well-defined stages. The complete operation is controlled by the Finite State Machine (FSM).

6.1 Card Insertion

The system starts in the idle state. When the start signal is activated, the controller checks the card information. The system supports three registered cards.

6.2 PIN Authentication

After detecting a valid card, the controller moves to the PIN verification stage. The PIN entered by the user is compared with the PIN stored for the corresponding card.

If the PIN is correct, the user is allowed to access the transaction menu.

6.3 Incorrect PIN Handling

If an incorrect PIN is entered, the attempt counter is increased. The system allows a maximum of three incorrect attempts.

After the third incorrect attempt, the corresponding card is blocked temporarily. A timer is used to control the temporary blocking period.

6.4 Transaction Menu

After successful PIN verification, the controller enters the menu state. The user can select the required operation.

The available operations are:

- Balance enquiry
- Cash withdrawal
- Deposit
- PIN change
- Card ejection

6.5 Balance Enquiry

When balance enquiry is selected, the controller enters the balance state. The current balance of the authenticated account is provided at the output.

6.6 Cash Withdrawal

When withdrawal is selected, the requested amount is checked against the account balance.

If sufficient balance is available, the requested amount is deducted from the account and the cash output is activated.

6.7 Deposit

When deposit is selected, the entered amount is added to the balance of the authenticated account. The updated balance is maintained by the controller.

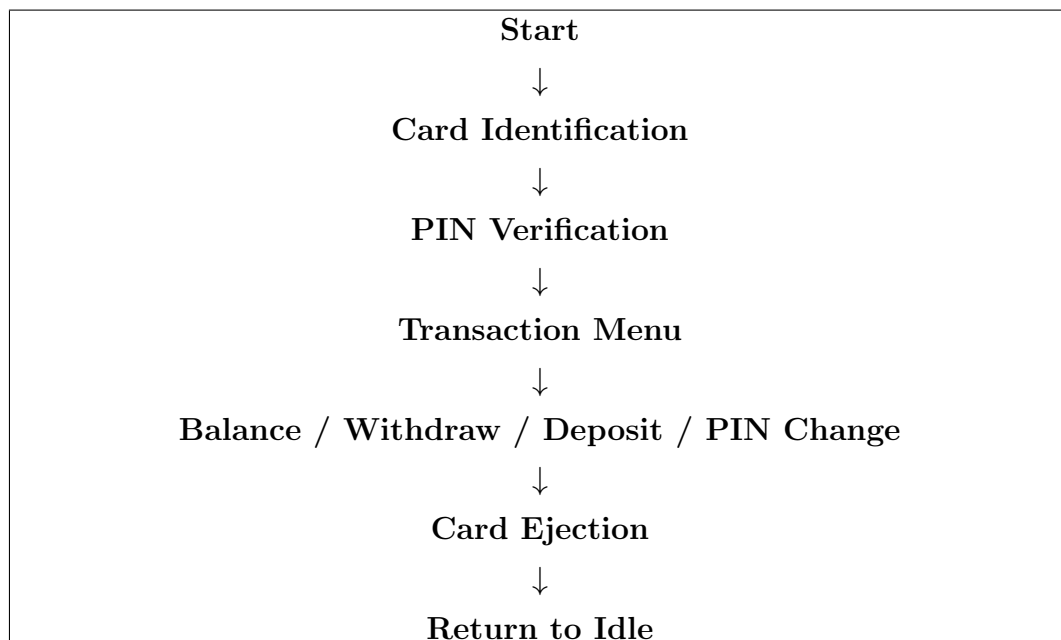
6.8 PIN Change

The authenticated user can select the PIN change operation. The new PIN is stored for the corresponding card.

6.9 Card Ejection

After completing the required operation, the user can select the eject option. The controller enters the card ejection state and then returns to the initial state.

ATM WORKING SEQUENCE



Operational Sequence of the ATM Controller

7. STATE-BASED CONTROL

The ATM controller uses a state-based control mechanism to organize the sequence of banking operations. The controller changes its state according to the inputs received from the user and the current operation being performed.

The states defined in the ATM design represent different stages of the transaction process.

7.1 Initial State

The controller begins in the **IDLE** state. This is the starting condition of the ATM system before a transaction begins.

7.2 Card Processing

When the transaction is started, the controller moves to the **CARD INSERTED** state. The card information is checked against the registered cards.

7.3 PIN Processing

The controller uses the **PIN CHECK** and **PIN ENTER** states for PIN authentication. The entered PIN is compared with the stored PIN associated with the selected card.

Incorrect PIN attempts are counted. After three incorrect attempts, the corresponding card is blocked temporarily.

7.4 Main Menu

After successful authentication, the controller reaches the **MENU** state. From this stage, the user can select the required banking operation.

7.5 Banking Operations

The controller provides separate states for the major operations:

- **BALANCE** – Used for balance enquiry.
- **WITHDRAW** – Used to process a withdrawal request.
- **DISPENSE CASH** – Used to activate the cash output after a successful withdrawal.
- **DEPOSIT** – Used to process a deposit.
- **PIN CHANGE** – Used to update the PIN.

7.6 Card Ejection

The design contains an **EJECT CARD** state. The **eject** signal is an input used to control the card-ejection operation; it should not be confused with the state itself.

8. WORKING FLOWCHART

The working flowchart represents the sequence of operations performed by the ATM controller. The system begins from the idle condition, identifies the card, verifies the PIN, and then provides access to the selected banking operation.

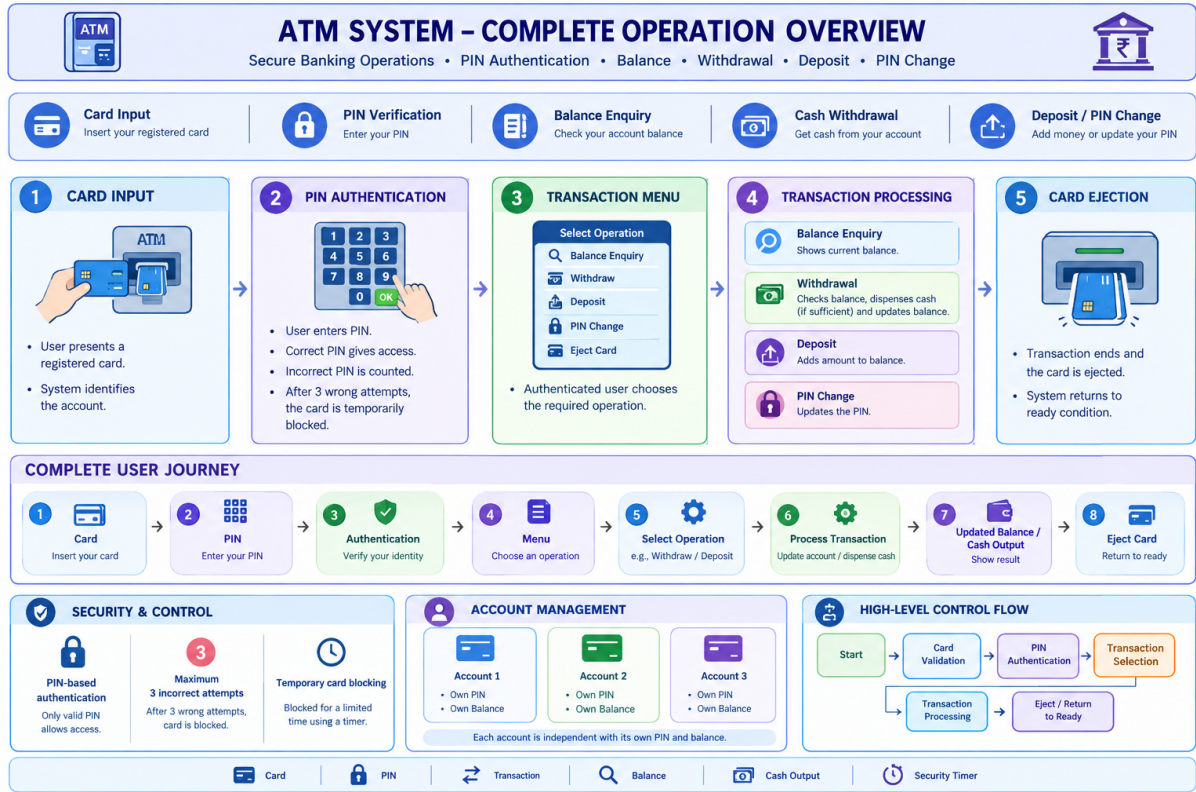


Figure 1: ATM WORKING FLOW

9. RTL DESIGN

The RTL design represents the hardware structure generated from the Verilog HDL description. It shows the synthesized digital logic and the connections between the different elements of the ATM controller.

Working Flowchart and RTL Design

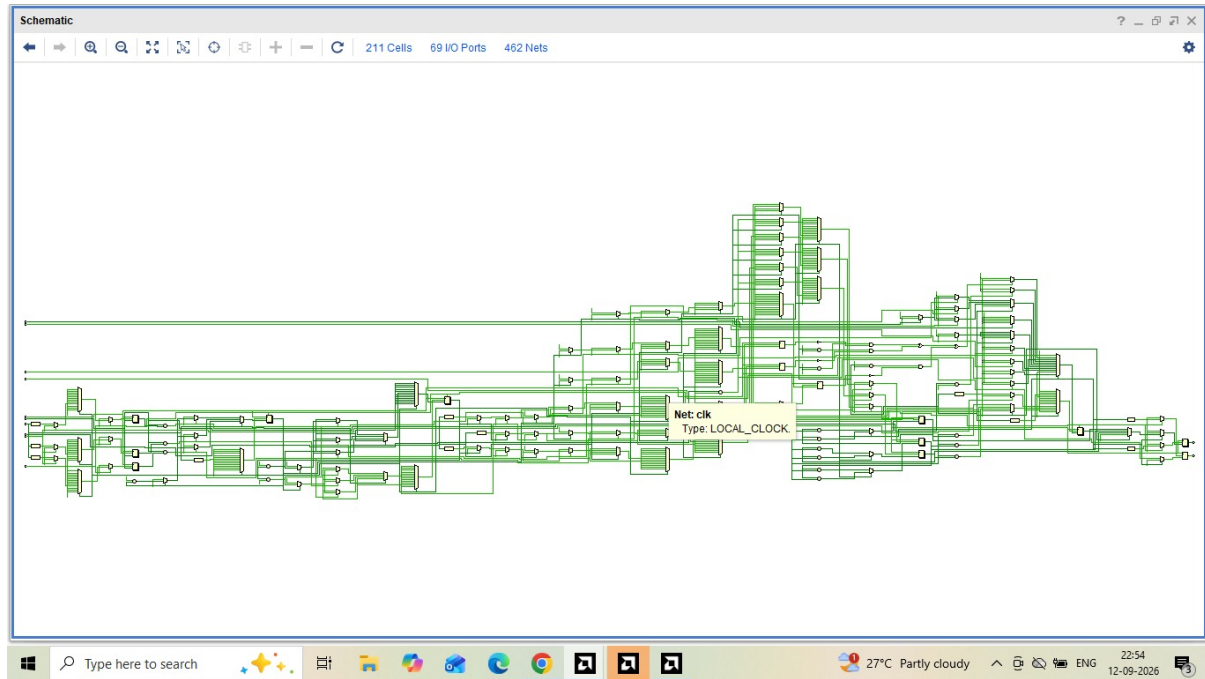


Figure 2: RTL DESIGN

10. SIMULATION AND RESULTS

The ATM controller is verified using a Verilog testbench. The simulation is used to verify the different operations of the ATM system, including card identification, PIN verification, balance enquiry, withdrawal, deposit, PIN change, incorrect PIN attempts, card blocking, and cash output.

The simulation waveform demonstrates the response of the ATM controller to different input conditions. The changes in the state, balance, cash output, and other control signals can be observed during the verification process.

Simulation Verification

- Card identification is verified for the registered cards.
- Correct PIN authentication is verified.
- Incorrect PIN attempts are counted.
- Card blocking after three incorrect attempts is verified.
- Balance enquiry is verified.
- Withdrawal operation is verified.
- Deposit operation is verified.
- PIN change operation is verified.
- Cash output during successful withdrawal is verified.

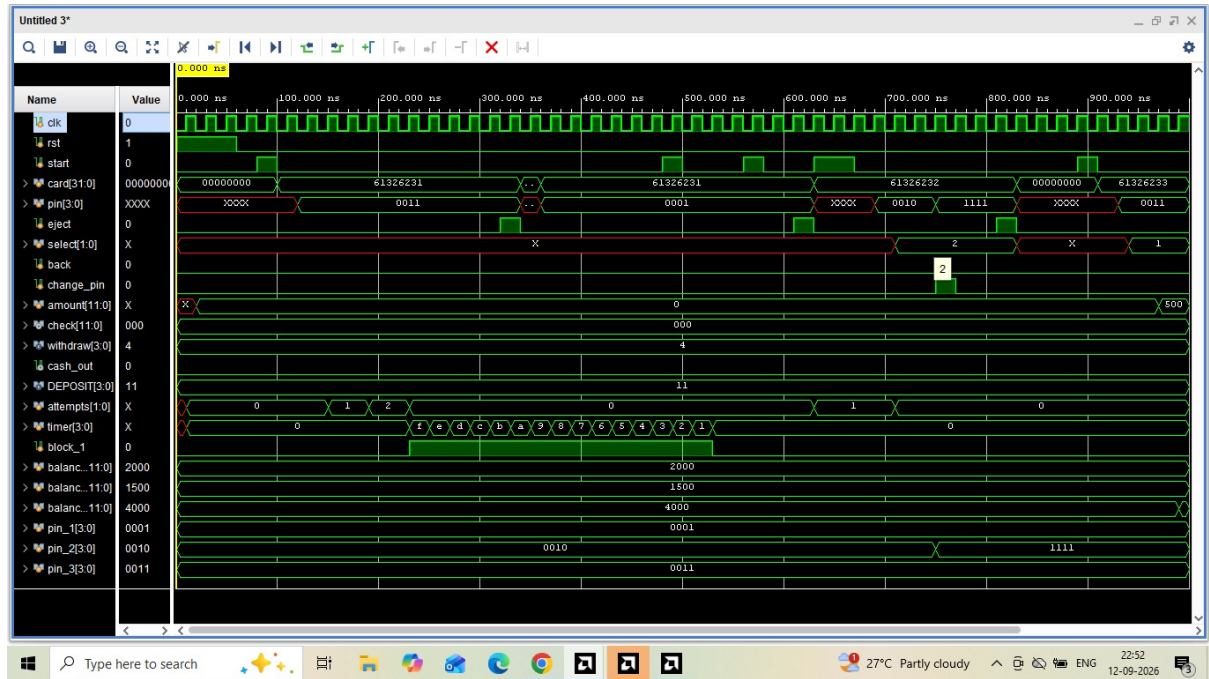


Figure 3: SIMULATION

- Card ejection operation is verified.

Simulation and Verification Results

11. APPLICATIONS

The ATM controller can be used as a basic model for understanding automated banking operations and digital authentication systems.

The major applications are:

- Educational Verilog HDL projects.
- Digital system design laboratory experiments.
- FSM-based controller demonstrations.
- Banking transaction control system studies.
- Digital authentication and security system demonstrations.
- RTL design and simulation learning.

12. ADVANTAGES

The major advantages of the ATM controller are:

- PIN-based authentication provides controlled access.
- Supports multiple registered accounts.
- Provides balance enquiry functionality.
- Supports withdrawal and deposit operations.
- Allows the user to change the PIN.
- Provides incorrect PIN attempt counting.
- Temporarily blocks the card after three incorrect attempts.
- Uses a timer for temporary blocking.
- Provides cash output for a successful withdrawal.
- The design can be verified through RTL simulation.
- The project demonstrates practical application of digital design concepts.

13. CONCLUSION

The ATM controller project successfully demonstrates the design of a basic automated banking system using Verilog HDL.

The system provides card identification, PIN authentication, balance enquiry, cash withdrawal, deposit, PIN change, and card ejection operations. Security is also considered through incorrect PIN attempt counting and temporary card blocking after three unsuccessful attempts.

The project demonstrates the practical use of digital logic, registers, counters, timers, and state-based control in a real-world application. The functionality of the design is verified through RTL simulation using a testbench.

Overall, the project provides a clear understanding of how a digital ATM controller can be designed, simulated, and verified using Verilog HDL.